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Assessing Tributary Effects on Sediment Composition in the Dam-Regulated Dixence River.

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Grande Dixence Dam. *Image taken from "https://www.valais.ch/en/explore/activities/other-summer-activities/dams-tour/grande-dixence"*

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Abstract

We investigate the influence of tributary inputs on sediment composition in the Dixence River downstream of the Grande Dixence Dam in Switzerland. Using high-resolution imagery and manual photo-sieving techniques, we analyzed sediment grain-size distributions (D_{50}, D_{65}, D_{84}) and sand content (PS) across 240 sampling points along a 12 km river reach. Our results reveal a pronounced downstream fining trend, with grain sizes decreasing and sand content increasing with distance from the dam, consistent with dam-induced sediment starvation. Tributary confluences exhibited heterogeneous effects on sediment composition, with some contributing coarser material (e.g., T1, T4) and others introducing finer sediments (e.g., T2, T5). The variability in tributary influence was linked to catchment characteristics such as lithology, slope, and area. Notably, the Sediment Sorting Index (SSI) consistently increased downstream of confluences, suggesting enhanced sediment organization. This study helped in identifying tributaries with significant impacts on riverbed composition (e.g., T1, T5) and provided a preliminary assessment of the sediment contribution from the studied tributaries. It also showed the complexity of the interactions between the tributaries and the river and a more detailed investigation for these tributaries is needed in order to be able to better predict their impact downstream.

Keywords: *Sediment dynamics; Downstream fining; Dam impacts; Tributary inputs; Alpine rivers; Grain-size distribution; Sediment management*

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1. Introduction

Bedload composition—defined by grain size, sorting, and lithologic diversity—is a critical but often overlooked control on river morphodynamics. Coarse sediment maintains channel stability (Misset et al., 2020; Surian, 2002), provides spawning substrate for aquatic biota (Merz et al., 2006; Haidvogl et al., 2015; Bombino et al., 2009), and mediates floodplain interactions (Fryirs et al., 2007; Misset et al., 2020). Yet, traditional sediment management strategies (e.g., mechanical replenishment, sediment bypass tunnels) frequently fail to replicate natural bedload regimes, as they are costly (Auel and Boes, 2012), spatially limited, and often disregard tributary inputs (Lane et al., 2022).

Dams are among the most widespread anthropogenic disturbances to riverine sediment dynamics, with over 50,000 large dams globally disrupting fluvial connectivity (Lehner et al., 2011) . While hydropower provides approximately 16% of the world's electricity (Pennsylvania State University, 2011), their sediment-trapping effects have profound ecological and geomorphological consequences downstream (Kondolf et al., 2014a), particularly through the trapping of coarse bed-load, which constitutes the structural foundation of river channels and thus compromising both the physical integrity and ecological function of river systems (Fryirs et al., 2007; Kondolf et al., 2014a). In alpine regions, where dams are densely concentrated, the retention of gravel- to cobble-sized sediment triggers channel incision, armoring, and habitat degradation (Lane et al., 2022; Kondolf, 1997), severing ecological connectivity and reducing geomorphic complexity (Kondolf et al., 2006; Lane et al., 2022). The geomorphic impacts of dams were first systematically documented by Williams and Wolman (1984) , who observed post-

impoundment channel narrowing, bed coarsening, and reduced sediment mobility in U.S. rivers. Subsequent studies revealed that coarse bedload is disproportionately trapped compared to suspended sediment, disrupting the balance between sediment supply and transport capacity (Brandt, 2000; Kondolf, 1997). In alpine rivers, where sediment fluxes are naturally high, dams exacerbate sediment disconnectivity, leading to downstream starvation (Kondolf et al., 2014a; Lane et al., 2022).

However, in an Alpine setting, unregulated tributaries may also contribute significant amounts of bedload-sized sediment and so compensate for sediment trapped behind dams (Lane et al., 2022; Milner et al., 2019). In natural gravel-bed rivers, research by Rice (1999), Rice (1998), and Rice and Church (1998) has shown that tributaries can disrupt downstream fining patterns—where grain size typically decreases with distance—by introducing coarse sediment at confluences. They introduced the concept of "sedimentary links", which are reaches between significant lateral sediment inputs where fining processes dominate, punctuated by abrupt grain-size increases at tributaries (Rice, 1999). In our case, given the dam-induced sediment starvation and altered hydrological conditions, the tributary influence on bedload composition might be more pronounced. Reduced upstream sediment supply could render tributary inputs disproportionately more significant, leading to pronounced local increases in grain size and improved sorting immediately downstream of confluences. This aligns with a geomorphological interpretation of the serial discontinuity concept (Lane et al., 2022), where tributaries may provide a lot of coarse sediments, leading to "patches of geomorphic recovery" and mitigating dam-related sediment disconnection in Alpine rivers. While in dammed basins, tributaries can locally reactivate bedload transport, creating patches of geomorphic recovery downstream of confluences (St. Pierre Ostrander et al., 2024; Lane et al., 2022), their effectiveness hinges on sediment caliber, hydrological regime, and basin lithology (St. Pierre Ostrander et al., 2024; Li et al., 2019; Nickolas et al., 2017) , factors that remain poorly quantified in alpine contexts.

The goal of this thesis is to assess whether and how tributary inputs influence sediment composition in an alpine river reach downstream of a major dam. Specifically, it addresses the following research questions: how does sediment grain size and sorting vary with distance from the Grande Dixence Dam? To what extent do tributary confluences alter local sediment composition (D50, D65, D84, PS)? What physical or geomorphological characteristics explain the variability in tributary influence across sites? We hypothesize that while the dam imposes a strong longitudinal fining trend, tributaries will locally disrupt this pattern—particularly through short-range increases in grain size and decreases in sorting quality downstream of confluences. To test these hypotheses, sediment metrics (D50, D65, D84, and PS) were extracted from 240 georeferenced riverbed images along a 12 km reach below the Grande Dixence Dam. Using manual photo-sieving, spatial modeling, and statistical analysis, we evaluated sediment trends in relation to both dam distance and tributary proximity.

2. Methods

2.1 Study Area

This study was conducted along the Dixence River, the main outflow of the Grande Dixence Dam located in the Val d'Hérémence in the canton of Valais, Switzerland. The Grande Dixence is the tallest gravity dam in the world, standing at 285 meters. Construction began in 1951 and was completed and commissioned in 1961 (Grande Dixence SA, 2025a). For reference as to how big this dam is, it weighs more than the Great Pyramid of Giza and has enough concrete inside it to build a 1.5 meters high wall across the equator of the earth (Grande Dixence SA, 2025a). It serves as a major hydroelectric infrastructure, feeding four pumping stations and three production plants, producing 2 billion kWh every year, accounting for 20% of Switzerland's storage energy and supplying electricity to over 500,000 households (Grande Dixence SA, 2025b). The dam retains glacial meltwater from multiple alpine valleys, significantly reducing the downstream transport of coarse sediments. This makes the area particularly well-suited for examining the impacts of sediment trapping on riverbed composition. Additionally, the Dixence River receives input from numerous tributaries along its course, creating heterogeneous flow and sediment conditions ideal for spatial analysis.

Based on the lithology map in Appendix A, which was taken from map.geo.admin.ch, the river corridor in the study area is primarily underlain by a homogeneous

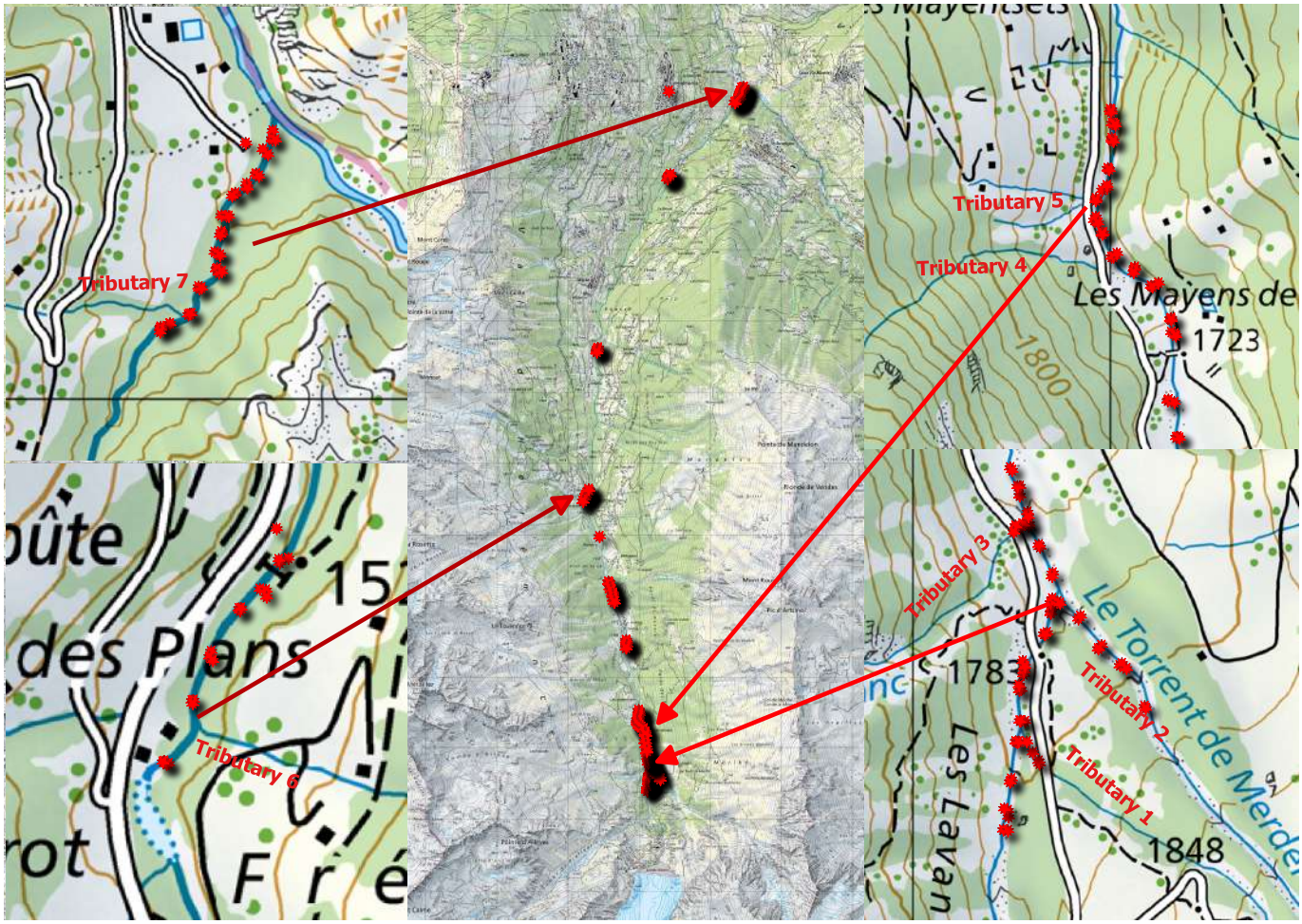


Figure 1. Area of Interest. *In the middle we can see the full study area map. On the left and right sides we have zoomed in pictures on the area of interest with each red star representing a georefrenced image. This map was taken from Swisstopo, added and edited in QGIS.*

sequence of clays, silts, and sands. This consistent sedimentary composition dominates nearly the entire river section, except for a localized downstream zone where the substrate shifts to dolomites and quartzites, forming a distinct lithological transition. In contrast, the broader catchment area exhibits greater lithological heterogeneity. The dominant geological units consist of mica schists and gneisses, interspersed with discontinuous occurrences of blocks, gravels, limestones, partially marly formations, quartzites, and dolomites.

Sediment composition was assessed through high-resolution imagery of the riverbed rather than traditional sieving methods. This approach enabled efficient and non-invasive data collection of a large sample size, with subsequent image processing and particle size analysis performed in MATLAB. Fieldwork was conducted over three consecutive days. Each day focused on a distinct segment of the river, moving progressively upstream from the confluence between the Dixence river and the Borgne river (46.18007°N, 7.41889°E) toward the dam. Equipment used included a "1x1m sampling grid", a GPS "Garmin eTrex 20x" unit, and a smartphone camera. At each sampling site, two or three lateral photos were taken depending on river width. Sampling locations were spaced 25 to 40 meters apart along the river's longitudinal profile. This sampling distance provides sufficient detail to capture changes along the river's length while keeping data collection efficient and manageable. Some segments were excluded due to excessive depth, flow velocity, or physical inaccessibility. In total, 240 sampling points were recorded, with the highest spatial density concentrated within the first 6 kilometers downstream of the dam. The sampled reach covered approximately 12 kilometers of the river, extending from the base of the dam to

lower elevation zones.

2.2 Extraction of Sediment Metrics

To quantify grain-size distributions on the riverbed surface, a manual photo-sieving approach was used, based on high-resolution images taken with a 12 MPx rear camera (Samsung Galaxy S24 Ultra). Sampling areas were defined using a 1 m $\times$ 1 m wooden quadrat frame, with measuring tape along all sides for precise scaling. Images were processed using a custom `LaneGrain_v1.m` MATLAB script, structured in three steps:

1. **Image Calibration:** A 10 cm reference segment was manually selected on each image to calibrate scale, using the `scale.m` function. The scale factor (s) converting pixels to physical units was computed as:
$$s = \frac{0.1}{\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}},$$
where (x_1, y_1) and (x_2, y_2) are pixel coordinates of the reference points.

2. **Random Particle Selection:** A set of $n_p = 100$ particles was randomly selected within each image using uniform sampling. Random coordinates (x_r, y_r) ensured representative coverage across the quadrat.

3. **Grain Measurement:** Using `grain.m`, the intermediate (*b*-) axis of each particle was manually measured by clicking its endpoints. Grain size was computed as:
$$g_{\text{size}} = s \cdot \sqrt{(x_2^{(i)} - x_1^{(i)})^2 + (y_2^{(i)} - y_1^{(i)})^2},$$
where $(x_1^{(i)}, y_1^{(i)})$ and $(x_2^{(i)}, y_2^{(i)})$ are the i-th grain's coordinates.

From these measurements, percentile grain sizes (D_{50}, D_{65}, D_{84}) and Percentage of Sand (PS). Then using (D_{50}) and (D_{84}) we calculated a simplified Sediment Sorting Index (SSI).

2.3 Distance Metrics

To quantify the spatial relationship between sampling points and key fluvial features, two sets of distance metrics were calculated: (1) the distance from each point to the Grande Dixence Dam, and (2) distance relative to nearby tributaries.

The distance from the dam to each sampling point was computed using Python. The dam coordinates were defined as 46.08056°N, 7.40389°E. Two GPX files containing the river's waypoint data were parsed and merged using the gpxpy library. All waypoint coordinates were extracted, and geodesic distances from the dam were calculated using the geopy library. The output—comprising waypoint names, coordinates, and distances in kilometers—was organized into a pandas DataFrame and exported as a CSV file for further analysis.

Tributary-related distances were calculated manually using Google My Maps. For every sampling point, the distance along the river to the closest upstream tributary was measured by following the river's path—not a straight line. This provided a continuous upstream distance metric for all points. In addition, a tributary-centered distance system was developed. For each tributary, all sampling points located approximately 200 meters upstream and downstream were identified. Distances from these points to the tributary were measured along the river's course. Points upstream were assigned negative values, and those downstream were assigned positive values, with the tributary defined as position zero. This allowed for a relative distance metric centered on each tributary, enabling analysis of sediment changes around confluences. All values were recorded in an Excel spreadsheet and later exported to CSV format for further processing in R.
The concavity, slope, and catchment area of each tributary were determined using `www.map.geo.admin.ch`. A line was drawn along the path of each tributary to extract elevation data. Concavity was assessed from the elevation profile shape. Slope was calculated as the difference between the highest and lowest elevations divided by the tributary length. Catchment areas were obtained using the website's catchment filter tool which gives the catchment area simply by selecting the relevant catchment.

2.4 Statistical Analysis

All statistical analyses were conducted using R (version 4.4.1). To investigate spatial trends in sediment composition, we modeled the relationship between each sediment metric (D50, D65, D84, PS) and distance predictors. These included both the distance from the Grande Dixence Dam and the relative distance to tributaries. To explore potential non-linear relationships, we fit stan-

dardized linear models using polynomial (second- and third-order) and log-transformed distance terms. To evaluate the effect of individual tributary confluences, we implemented a "before-and-after" comparison approach. For each tributary (T1–T7), sediment metrics within upstream and downstream were compared. We categorized sampling points into binary groups ("Before 0" and "After 0") relative to the tributary location. For each group comparison, the normality of data distributions was first tested using the Shapiro–Wilk test. Depending on the outcome, either independent-samples t-tests or Mann–Whitney U tests were applied to assess differences in median values.

2.5 Figures and Visualization

The R package ggplot2 (version 3.4.4) was used for visualization. The R package viridis (version 0.6.5) was used to specify the color palette.

2.6 Code and Availability

The code used to generate the simulated data and produce the figures, is available in the GitHub repository associated with this project link.

3. Results

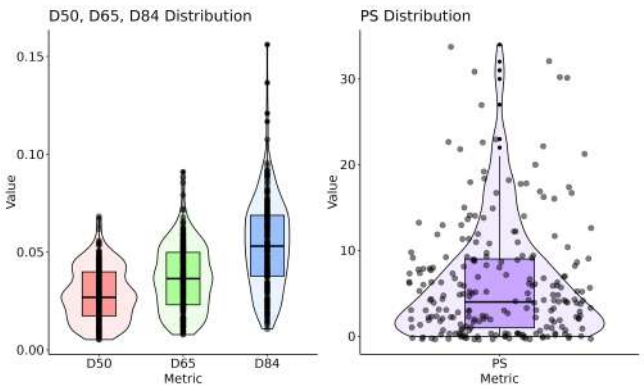


Figure 2. Distribution of Sediment Metrics (D50, D65, D84, and PS).

First, the distribution of the four extracted metrics—D_{50}, D_{65}, D_{84}, and PS (Percentage of Sand)—was examined to assess overall sediment characteristics and verify internal consistency. Summary statistics are presented in Table 1, while the corresponding distributions are visualized in Figure 2. As expected, the three percentile metrics display a progressive increase in central tendency, with mean values

Table 1. Descriptive statistics of sediment metrics (D50, D65, D84, PS). Summary statistics corresponding to the violin plots presented in Figure 1.

Metric	Min	Q1	Median	Q3	Max
D50	0.010	0.020	0.025	0.030	0.059
D65	0.013	0.025	0.032	0.041	0.089
D84	0.018	0.037	0.051	0.067	0.155
PS	0.00	2.33	4.12	8.36	34.26

of 0.03m for D_{50}, 0.04m for D_{65}, and 0.05m for D_{84}[1]. The medians mirror these values, indicating relatively symmetric distributions. All three variables exhibit low variability, with standard deviations ranging from 0.01 to 0.02m and limited ranges (0.06–0.15 m). The violin plots further confirm these trends, showing moderately right-skewed distributions, particularly for D_{84}, due to the presence of larger particles in the upper tail. In contrast, PS exhibits a much wider and more irregular distribution, with values ranging from 0% to 34% and a standard deviation of 6.91%. The median PS is 4%, suggesting that sand-sized particles (<2 mm) are generally a minor component of the riverbed surface. However, several high outliers indicate localized zones with substantially higher fine material content. The variability observed in PS reflects the substantial variation in percentage of sand between the study sites.

The relationship between sediment metrics and the logarithmic longitudinal distance from the "Grande Dixence Dam" was assessed to examine spatial patterns in grain size and sand content Figure 3. A clear negative trend was observed for all three grain size percentiles. D_{50}, D_{65}, and D_{84} values consistently decreased with increasing distance from the dam, indicating progressive downstream fining within the first 12 km. Conversely, PS (Percentage of Sand) increased with distance, suggesting that finer material becomes more prevalent further downstream. To quantify these trends, linear models were fitted using standardized sediment metrics as response variables and distance as the predictor. The standardized coefficients Figure 4 confirm these observations: distance from the dam had a strong negative effect on D_{50}, D_{65}, and D_{84} (stan-

[1] Overall, the consistent progression of the percentile metrics supports the reliability of the measurement approach and serves as a quality control check, in simpler terms, if we saw that D_{84} median was lower than that of D_{65} for example, it means some mistake was done during the data processing.

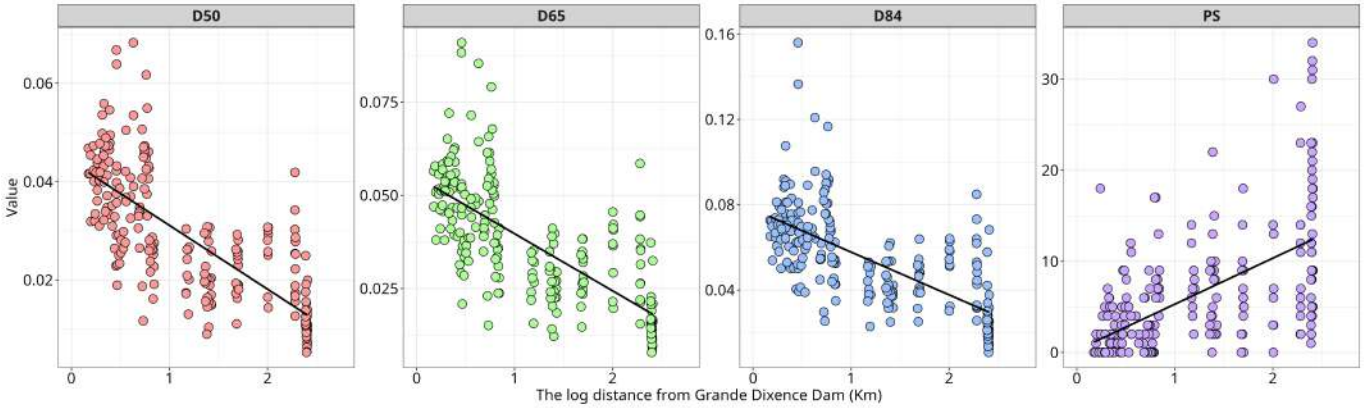


Figure 3. Relationship Between Metrics and Distance from Grande Dixence Dam. *Scatterplots with regression lines, showing the relationship between D50, D65, D84, and PS metrics and the log distance from the Grande Dixence Dam. The R^2 values can be found in Figure 4*

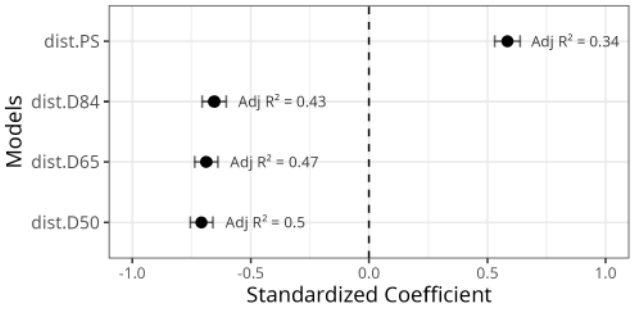


Figure 4. Standardized Coefficients for Metrics with log Distance from Grande Dixence Dam. *A plot of standardized coefficients with error bars for D50, D65, D84, and PS, capturing the influence of log distance from the Grande Dixence Dam.*

dardized β values ranging from approximately -0.65 to -0.75), and a strong positive effect on PS ($\beta \approx 0.55$). All relationships were statistically significant, and error bars around the coefficients were narrow, indicating high confidence in the effect estimates. These results are consistent with the expected influence of sediment trapping by the dam, where coarse material is retained near the outlet, and finer sediments dominate farther downstream. The sharp gradient in sediment properties within a short downstream distance highlights the localized geomorphic impact of the dam on riverbed composition.

To evaluate the robustness and potential non-linearity in the effect of distance from the dam on sediment metrics, additional linear models were tested using polynomial transformations of distance. These included distance (`dist`), squared distance (`dist2`), cubed distance (`dist3`), and log-transformed distance (`log_dist`). Standardized coefficients for all models

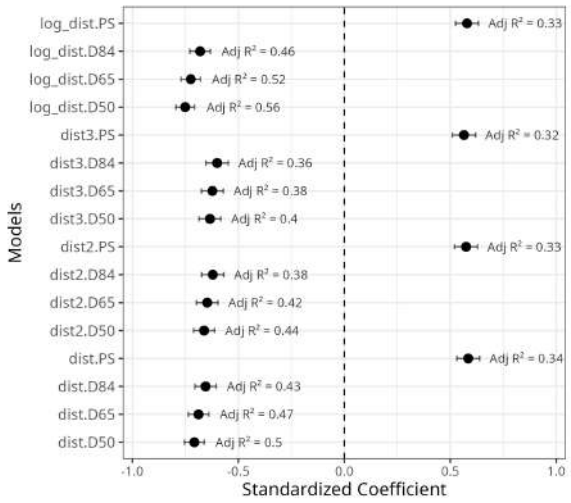


Figure 5. Standardized Coefficients Across Multiple Models for Grande Dixence Dam Distance. *A comprehensive comparison of standardized coefficients with error bars for various modeling approaches, examining the impact of distance from the Grande Dixence Dam on D50, D65, D84, and PS metrics.*

are shown in Figure 5. The results consistently show strong negative relationships between distance and the three grain size percentiles (D_{50}, D_{65}, and D_{84}), with standardized coefficients ranging from approximately -0.5 to -0.8 across all transformations. Conversely, PS (Percentage of Sand) displayed strong positive associations with distance in all models, with standardized coefficients around $+0.4$ to $+0.6$. While the form of the models differed, the positive trend remained consistent, indicating that the observed spatial patterns—finer material accumulating downstream and coarser grains concentrated near the dam—are robust to differences in model specification. The narrow confidence intervals across all coefficients further support the reliability of these results.

The influence of distance from the nearest upstream tributarie on sediment metrics was also assessed. As shown in Figure 6, grain size percentiles (D_{50}, D_{65}, and D_{84}) exhibited weak positive trends with increasing distance from the nearest tributary, while PS (Percentage of Sand) showed a weak negative trend. These relationships were visually subtle and exhibited high scatter, indicating a weaker and less consistent pattern compared to the trends observed with distance from the dam. To quantify these trends, standardized linear models were fitted using tributary distance as the predictor variable Figure 7. Results showed small positive coefficients for the grain size percentiles (approximately $+0.1$), and a small negative coefficient for PS (approximately -0.15). However, confidence intervals around all estimates were relatively wide and overlapped zero in most cases, suggesting that the observed relationships are weak and potentially not statistically significant. These results indicate that proximity to tributaries may have only a minor influence on surface sediment size distribution at the reach scale examined. While tributary inputs can locally affect sediment composition, their impact appears secondary compared to the dominant influence of the dam.

To further explore the sensitivity of sediment metrics to tributary proximity, additional models were fitted using transformed versions of the tributary distance variable, including squared distance (`dist_trib2`), cubed distance (`dist_trib3`), and log-transformed distance (`log_dist_trib`). The standardized coefficients and associated error bars are shown in Figure 8. Overall, the effects of tributary distance on sediment metrics remained weak across all transformations. For the grain size percentiles (D_{50}, D_{65}, and D_{84}), the squared and cubed models produced small but consistently positive coefficients, while PS (Percentage of Sand) showed negative coefficients across the same models. Although these effects were modest in magnitude, their confidence intervals generally remained on one side of zero, suggesting weak but directionally consistent relationships. In contrast, the log-transformed models yielded larger absolute coefficients in some cases but exhibited wide confidence intervals that crossed zero for all metrics. This indicates that the log-transformed models introduce greater uncertainty and fail to provide a clear direction of effect. Therefore, while none of the models showed strong or statistically robust associations, the polynomial transformations (squared and cubed) offered more consistent patterns compared to the log-transformed version. These findings suggest that the influence of tributary proximity on surface sediment composition might be limited in magnitude and dependent on how distance is modeled. However, the observed directional consistency in polynomial models may still indicate subtle localized effects worth exploring further.

To explore potential interactions between dam proximity and tributary influence, sediment metrics were plotted against the logarithm of the distance from the Grande Dixence Dam, with points colored according to their distance from the nearest upstream tributary Figure 9. This dual-gradient visualization aimed to capture how sediment composition varies along the main longitudinal profile while accounting for lateral contributions from tributaries. Consistent with earlier findings, all three grain size percentiles (D_{50}, D_{65}, and D_{84}) exhibited a clear decreasing trend with increasing log-distance from the dam, indicating progressive downstream fining. Conversely, PS (Percentage of Sand) showed a gradual increase along the same gradient, consistent with the accumulation of finer material farther downstream. The color gradient revealed that sediment characteristics were also influenced by their proximity to the nearest upstream tributary. In the upper reaches, where dam influence is strongest, points closer to tributaries (indicated by warmer colors) tended to display slightly higher grain sizes and lower PS values compared to more distant sites. This pattern suggests that tributary inputs may locally enhance coarse material concentrations, potentially modulating

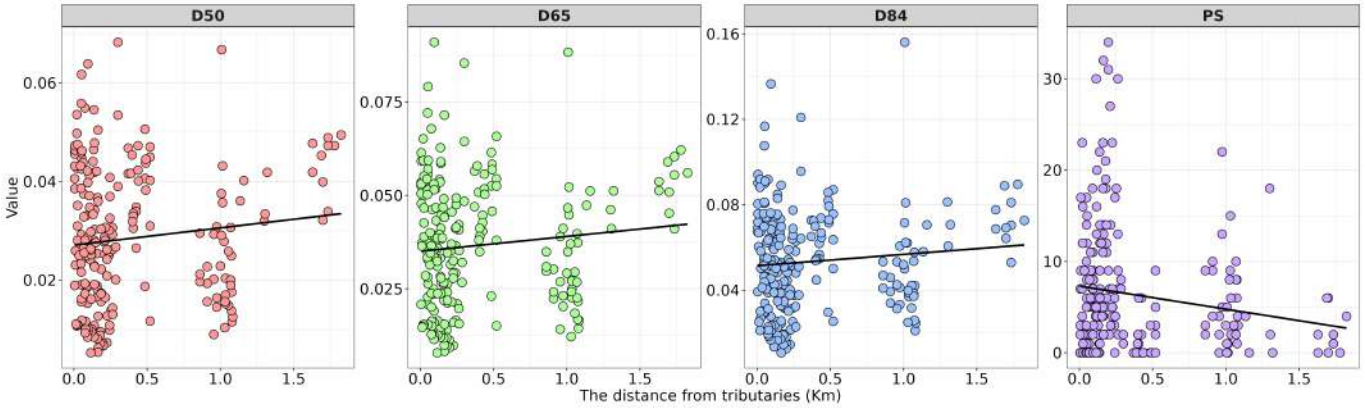


*Figure 6. **Relationship Between Metrics and Distance to Nearest Upstream Tributary.** Scatterplots showing the relationship between D50, D65, D84, and PS metrics and distance to the nearest upstream tributary measured by the following the river path and not a straight line. The R^2 values can be found in Figure 7.*

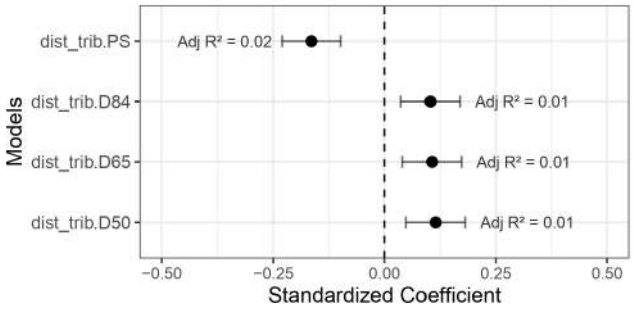


*Figure 7. **Standardized Coefficients for Metrics and Distance to Nearest Upstream Tributary.** A plot of standardized coefficients with error bars for D50, D65, D84, and PS metrics, indicating the influence of distance to nearest upstream tributary. Confidence intervals illustrate the precision of the estimates.*

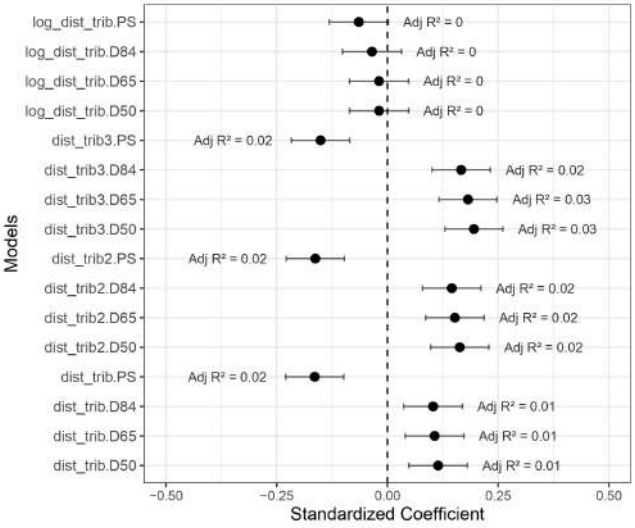


*Figure 8. **Standardized Coefficients Across Multiple Models for Distance to Nearest Upstream Tributary.** A comparative visualization of standardized coefficients with error bars for different modeling approaches, focusing on the impact of distance distance to nearest upstream tributary on D50, D65, D84, and PS metrics.*

the dominant fining signal imposed by the dam. Overall, the figure highlights how sediment properties are jointly structured by longitudinal distance from the dam and lateral contributions from tributaries. The use of a log-scale for distance enhances resolution near the dam and clarifies the nonlinear nature of the observed trends.

To investigate the local effect of tributary confluences on riverbed sediment characteristics, sampling points were grouped into upstream ("Before 0") and downstream ("After 0") categories relative to seven identified tributaries (T1–T7). For each tributary, the distributions of D_{50}, D_{65}, D_{84}, PS (Percentage of Sand), and SSI (Sediment Sorting Index) were compared using boxplots Figure 10. The results show considerable heterogeneity in sediment response across tributaries. For D_{50}, D_{65}, and D_{84}, several tributaries exhibited a decrease in grain sizes downstream (e.g., T2, T3 and T5), while others showed minimal or even opposite trends (e.g., T1 and T6). This suggests that the influence of tributaries on sediment coarseness is site-specific and potentially dependent on tributary discharge, sediment load, or geomorphological context. PS displayed a similarly variable response, decreasing downstream of some tributaries (e.g., T2, T4) but mostly increasing in the rest (e.g., T5, T6 and T7). The Sediment Sorting Index (SSI) is the only index that consistently increased after each tributary confluence. This indicates that there seems to be better sediment sorting downstream of tributary inputs. The suggests that either well-sorted material is being introduced or that downstream processes are contributing to a more homogeneous grain size distribution.

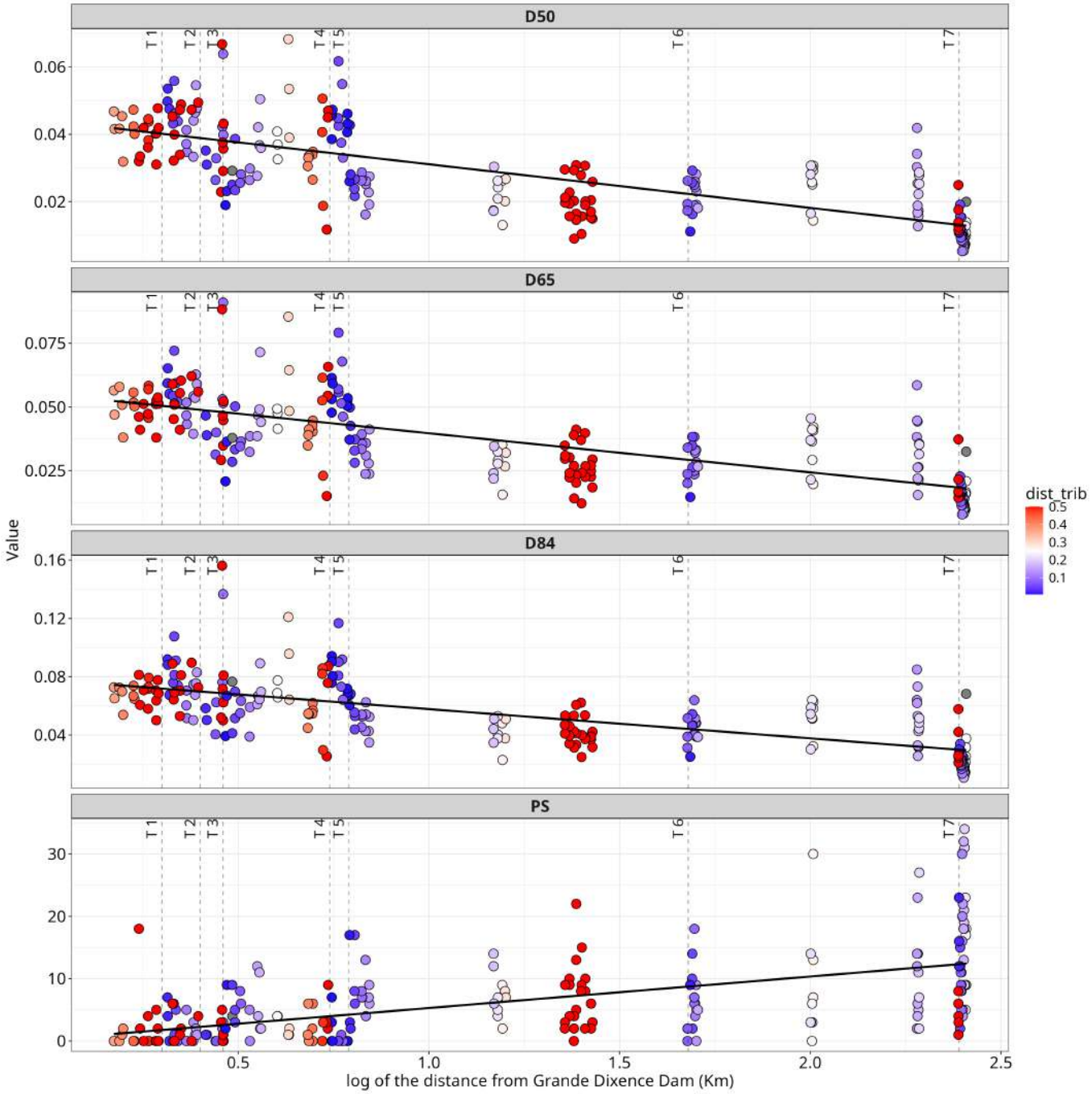


Figure 9. Influence of Combined Distances from Dam and Distance to Closest Upstream Tributary on Sediment Metrics. *Scatterplots illustrating the relationship between log-transformed distances (from the Grande Dixence Dam and tributaries) and sediment metrics (D50, D65, D84, and PS). The x-axis shows the natural log distance of each point from the dam, the color shows the distance from each point to the nearest upstream tributary.*

Table 2 summarizes the type of statistical test applied for each sediment variable across the seven tributary transects (T1–T7). The choice between independent samples *t*-tests and Mann–Whitney U tests (MWU) was based on the normality and variance assumptions assessed during preliminary analysis. Most sediment metrics, including D_{50}, D_{65}, D_{84}, and SSI, satisfied the assumptions for parametric testing in the majority of transects and were thus analyzed using *t*-tests. However, for PS, which frequently exhibited non-normal distributions, MWU was the preferred

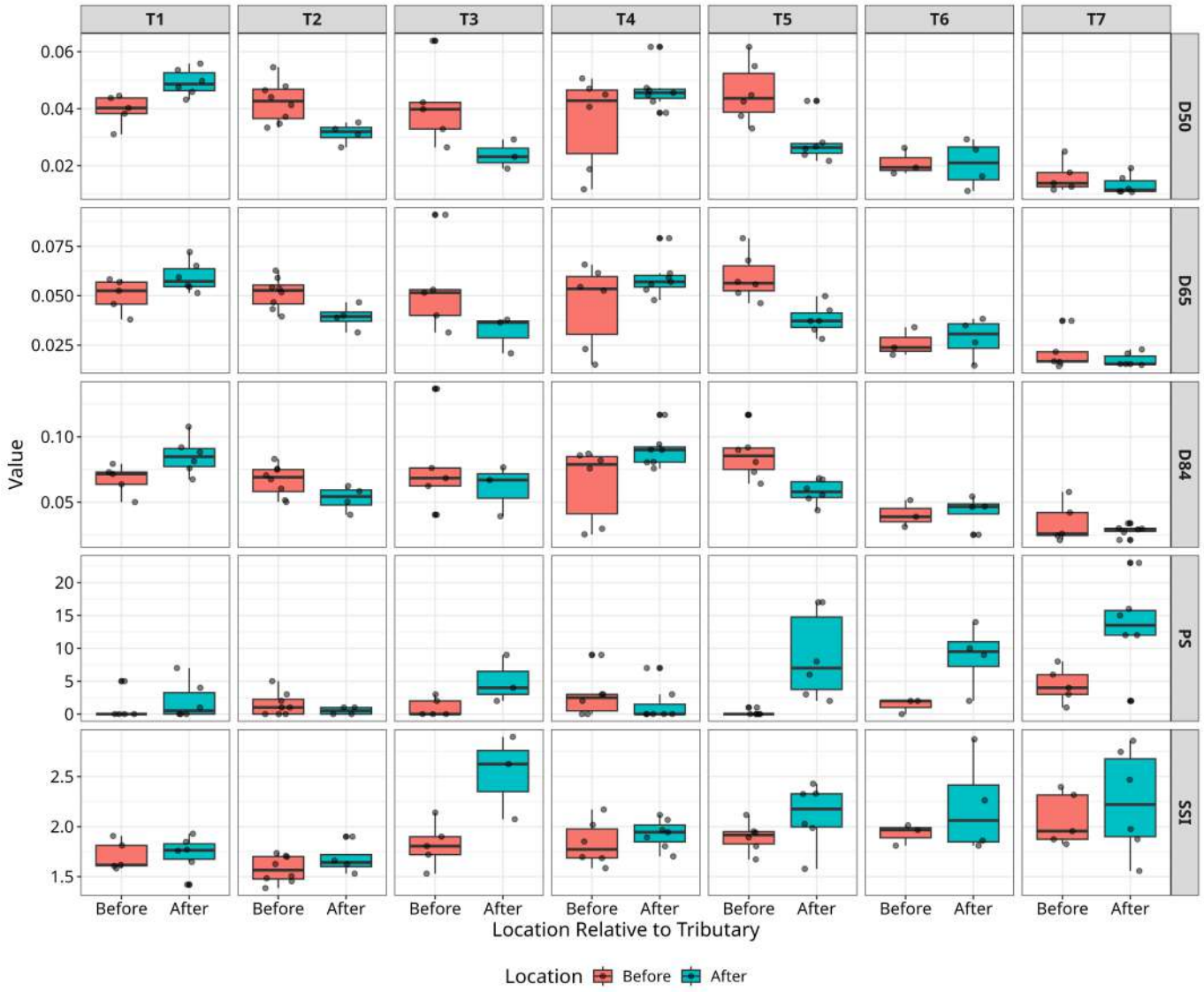


Figure 10. Boxplots with jittered points showing sediment characteristics (D50, D65, D84, PS, SSI) before and after tributary confluences across sites. *Columns represent tributaries; rows show variables.*

method for most transects (T1–T6). Additionally, MWU was used for D_{84} in T4 and for D_{50} in T5–T7, where normality assumptions were not met. These test selections ensured the robustness of comparisons between upstream and downstream conditions across each tributary. Table 2 also presents the associated p-values resulting from the statistical tests described above. Statistically significant differences (typically $p < 0.05$) were observed in multiple transects for various sediment variables. For D_{50}, significant differences were found in T1 ($p = 0.0136$) and T5 ($p = 0.0152$), indicating meaningful shifts in median grain size across these confluences. D_{65} also showed significant changes in T2 ($p = 0.0123$) and T5 ($p = 0.0052$), while D_{84} was significant in T1 ($p = 0.0429$) and T5 ($p = 0.0112$). PS exhibited significant differences across T5 ($p = 0.0036$) and T7 ($p = 0.0230$), aligning with visual trends noted earlier. No consistent patterns of statistical significance were observed for SSI, though marginally low p-values were noted in T3 and T5. These results indicate that sediment properties respond variably to tributary inputs, with some confluences inducing statistically significant changes in specific sediment metrics. T5 emerged as the most consistently significant location across multiple variables.

Table 2. Associated p-values for statistical tests on sediment variables per transect.

		T1	T2	T3	T4	T5	T6	T7
D50	**Test**	T-Test	T-Test	T-Test	T-Test	MWU	MWU	MWU
	P Val	**0.01363*****	0.0572**	0.05287**	0.16949*	**0.01515*****	0.9407	0.24675
D65	**Test**	T-Test	T-Test	T-Test	T-Test	T-Test	T-Test	T-Test
	P Val	0.09471**	**0.0123*****	0.11277*	0.1901*	**0.00522*****	0.71664	0.5368
D84	**Test**	T-Test	T-Test	MWU	T-Test	T-Test	T-Test	T-Test
	P Val	**0.04292*****	0.1441*	0.44911	0.1014*	**0.01117*****	0.77769	0.45922
PS	**Test**	MWU	MWU	MWU	MWU	MWU	MWU	MWU
	P Val	0.45923	0.41919	0.09086**	0.30959	**0.0036*****	0.09873**	**0.02302*****
SSI	**Test**	T-Test	T-Test	T-Test	T-Test	T-Test	T-Test	T-Test
	P Val	0.81759	0.4163	0.08128**	0.40305	0.17364*	0.35379	0.49803

Legend: **** Significant; ** Marginal; * Weak trend; No mark: Not significant*

Table 3. Physical characteristics of each transect (T1–T7), including slope, concavity type, and presence of sediment variables (flipped and color-coded).

	T1	T2	T3	T4	T5	T6	T7
D50	+	−	−	+	−	+	−
D84	+	−	−	+	−	+	−
PS	+	−	+	−	+	+	+
SSI	+	+	+	+	+	+	+
Concavity	Concave	Concave	Convex	Convex	Convex	Convex	Concave
Slope	0.3711	0.3597	0.4397	0.4210	0.4341	0.3486	0.4182
Catchment Area [km^2]	51.81	2.6	0.61	3.5	0.53	0.24	0.9

To explore potential drivers of sediment changes across tributary confluences, the direction of change in sediment metrics before and after each tributary was compared with local physical characteristics Table 3. Changes in D_{50}, D_{84}, PS, and the Sediment Sorting Index (SSI) were assigned either "+" or "−" based on whether the metric increased or decreased downstream of each tributary, as previously shown in Figure 10. Overall, there was no consistent relationship between concavity type (concave vs. convex) and the direction of sediment change. Both convex and concave profiles were associated with increases or decreases in grain size and PS, suggesting that channel curvature alone does not reliably predict sedimentary shifts. However, some patterns emerged when considering slope and catchment area. Tributaries with larger catchment areas (e.g., T1: 51.81 km^2) tended to show more consistent increases in sediment variables, possibly due to higher sediment supply. In contrast, tributaries with minimal catchment areas (e.g., T6: 0.24 km^2) and the lowest slope (0.3486) showed minimal to no increase in sediment metrics, and in some cases even a decrease. This suggests that the geomorphic and hydrological influence of smaller tributaries may be insufficient to produce detectable downstream changes in sediment composition. This just goes to show how complicated predicting the effect of tributaries can be.

4. Discussion

4.1 Downstream Fining and Dam-Induced Sediment Starvation

Grain size metrics (D_{50}, D_{65}, and D_{84}) revealed a pronounced downstream fining trend throughout the 12 km river reach studied below the Grande Dixence Dam. This progressive reduction in median and coarse grain sizes aligns with established patterns in dam-impacted fluvial systems, where sediment supply truncation leads to selective transport and downstream fining (Brummer and Montgomery, 2003; Kondolf et al., 2014b). The steep negative standardized coefficients across multiple regression models (linear and polynomial) underscore the robustness of this trend, reinforcing the dominance of dam-related sediment starvation near the outlet. Concurrently, the positive correlation between the percentage of sand (PS) and downstream distance further supports the hypothesis of progressive fine-material accumulation—a well-documented response to flow regulation and sediment trapping (Fuller et al., 2016; Petts, 2009). These collective trends reflect a classic post-dam adjustment, wherein the proximal reach exhibits sediment depletion, while finer fractions dominate farther downstream due to winnowing and reduced coarse sediment inputs (Brandt, 2000; Katano et al., 2021).

4.2 Tributary Influence on Main Channel Sediment Composition

In contrast to the relatively strong dam-proximity trends, Figure 6 showed that grain size metrics exhibited only weak correlations with distance from the nearest upstream tributary with considerable scatter, indicating that tributary proximity alone cannot explain the observed sediment size variations. This lack of a clear trend is largely due to the analysis method, which plots each sample by its distance to the nearest tributary upstream regardless of each tributary's special characteristics, thereby aggregating the diverse and site-specific influences of multiple confluences into a single relationship (Singer, 2008). Tributaries differ greatly in their sediment supply and impact on the main channel depending on factors such as their catchment area, sediment load, grain size characteristics, and hydrologic regime (Benda et al., 2005; Smažák and Galia, 2021). Larger or more energetic tributaries can introduce substantial coarse sediment that creates pronounced local changes in bed texture, whereas smaller tributaries or those delivering mainly fine material often produce negligible downstream effect (Benda et al., 2005; Singer, 2008). Consequently, analyzing all tributaries together is misleading, as it disregards and masks these individual differences and leads to high variability in the data. This is why tributary-specific analyses are necessary to discern the true extent of confluence impacts on main-channel sediment composition (Smažák and Galia, 2021).

As expected and mentioned in the previous section, the high scatter and low correlation that we saw when we stacked all the tributaries together were indeed caused by the fact that each tributary had a different effect on the main river. This is evident in the relationship analysis of individual tributary confluences in Figure 10. As observed in Table 2, several results are not statistically significant, so we will base our analysis on the significant findings. The significant increase in D_{50} and D_{84} downstream of Tributary T1 suggests that this confluence contributes coarse sediment to the main channel or enhances hydraulic conditions favoring the retention of coarser grains. Such coarsening aligns with studies documenting tributary-derived bedload inputs in mountainous systems (Benda et al., 2004). Tributary T1 exhibited the clearest coarsening signal, likely linked to enhanced bedload transport or reduced fine sediment supply downstream, a pattern resembling sediment armoring processes observed in regulated rivers (Kondolf et al., 2014a). A similar but much weaker coarsening effect was noted below Tributary T4 ($\uparrow D_{50}$, $\uparrow D_{84}$), suggesting a limited geomorphic influence, possibly due to lower sediment yield or confluence geometry minimizing sediment mixing (St. Pierre Ostrander et al., 2024). In contrast, Tributaries T2, T3, and T5 contributed to fining of the main-channel bedload, shown by declines in D_{50} and/or increases in percent sand (PS). Tributary T2 displayed modest declines in D_{50} and D_{65}, whereas T3 showed a decrease in D_{50} accompanied by increases in PS and SSI, pointing to localized inputs of finer, well-sorted sediments typical of low-energy confluences (Best, 1988). Among all tributaries, however, T5 had by far the most pronounced impact: all sediment parameters shifted significantly ($p < 0.05$), with decreases in D_{50} and D_{84} and sharp increases in PS and SSI. This highlights T5's dominant role as a major source of fine sediment, likely linked to erodible banks or urban

runoff (Allan, 2004). Such a strong tributary signal aligns with observations of tributaries functioning as key sediment hotspots in regulated river systems (Kondolf et al., 2014a; Li et al., 2023; Lane et al., 2022). Further downstream, PS increases below T6 and T7 continue to emphasize the cumulative role of tributaries in delivering fine material, likely sourced from easily erodible catchment surfaces (Allan, 2004; Li et al., 2019).

Although most SSI comparisons across tributary transects did not yield statistically significant p-values, the fact that all showed an increase downstream suggests a consistent underlying process may be at play. This uniform trend could point to subtle sediment filtering or hydraulic sorting effects at confluences, where grain-size distributions become slightly better organized despite minimal changes in D_{50} or D_{84} (Rice et al., 2006). Similar processes have been proposed in Alpine systems where tributary inputs modify sediment organization without necessarily altering median grain sizes (Benda et al., 2004; Zimmerman et al., 2010). While further investigation is needed to confirm these mechanisms, the observed SSI pattern highlights the potential for confluences to influence sediment structure in ways that traditional metrics may overlook (Czuba and Foufoula-Georgiou, 2015).

4.3 Influence of Catchment Lithology on Tributary Sediment Inputs

To further investigate the spatial variability of sediment properties across tributary confluences, the lithological characteristics of each tributary catchment were examined using the detailed geological datasets available on map.geo.admin.ch. A lithological map of the study area, presented in Appendix A, revealed that the tributaries feeding into the Dixence River traverse a range of geological substrates, including zones dominated by mica schists, gneiss, marly shales, calcareous phyllites, and deposits of clay, silt, and sand. The main channel itself flows predominantly through deposits of clay, silt, and sand, while the surrounding catchments are largely composed of mica schists and gneiss. Notably, the second tributary also originates in areas dominated by mica schists and gneiss before crossing finer lithologies downstream. These lithological units are theoretically expected to influence sediment caliber and composition by affecting erosion potential, weathering rates, and the dominant particle sizes delivered to the mainstem, with resistant lithologies producing coarser sediment and friable lithologies yielding finer grains (Roda-Boluda et al., 2018). While no fully systematic relationship was observed across all tributaries, a general pattern emerged: tributaries originating in coarser lithologies (mica schists and gneiss) tended to correspond with coarser sediment downstream of their confluences, whereas tributaries draining finer lithologies (marly shales, calcareous phyllites, clay, silt, and sand deposits) were generally associated with a fining of the mainstem sediment after the confluence. This observation suggests that lithology may influence sediment caliber at the confluence scale, although the signal appears subtle and likely interacts with other factors such as catchment area, slope, hydrological regime, or localized sediment storage. Consequently, while lithology provides important background control consistent with previous findings, it alone does not fully explain the heterogeneity observed in confluence-scale sediment responses. However, it is important to not that these results are based on a limited sample size and a much larger sample might be necessary.

4.4 Limitations and Opportunities for Future Research

This study encountered some limitations that should be acknowledged. Firstly, while the data resolution was generally good, certain sections of the alpine dammed river and its tributaries could not be surveyed due to inaccessibility or safety risks. As a result, not all tributaries were included, potentially limiting the comprehensiveness and generalizability of the findings. Secondly, an unforeseen challenge arose from the use of modern smartphones for imaging. These devices automatically pre- and post-process images, which complicated accurate identification and analysis of the river bed because of the water layer smoothing, which reduced the reliability of the visual data collected.

To address these limitations, several strategies are proposed for future research. Expanding the scope to include additional alpine dammed rivers would strengthen the generalizability of the findings and provide a broader understanding of these systems. Additionally, investing in a high-quality camera capable of capturing raw, unprocessed images is recommended to improve the accuracy of river bed analysis and enhance data quality. Furthermore, conducting the study a few times per year, over multiple years, would help us in examine temporal changes in the river systems. This approach would allow for the identification of trends,

variations, or stability in the characteristics of alpine dammed rivers, which could help in identifying the effects of rain, snow melt, etc ... These could prove of great importance and impact on the river bed sediment composition.

5. Conclusion

This study provides a basic analysis of the spatial variability in sediment composition along a 12 km reach of the Dixence River downstream of the Grande Dixence Dam, with a focus on the influence of tributary inputs. The results showed a pronounced downstream fining trend, characterized by decreasing grain sizes (D50, D65, D84) and increasing sand content (PS) with distance from the dam. This pattern aligns with established literature on dam-induced sediment starvation, where the trapping of coarse bedload disrupts natural sediment regimes and leads to progressive fining downstream. The strong negative correlations between grain size percentiles and dam proximity, coupled with the positive relationship between PS and downstream distance, underscore the dominant role of the dam in shaping sediment dynamics. Tributary confluences exhibited heterogeneous effects on sediment composition, with some tributaries (e.g., T1 and T4) contributing coarser material and others (e.g., T5 and T7) introducing finer sediments. The variability in tributary influence was linked to catchment characteristics such as lithology, slope, and area, with larger tributaries generally exerting more detectable impacts. Notably, the Sediment Sorting Index (SSI) consistently increased downstream of confluences, suggesting that tributary inputs may enhance sediment organization even when grain size changes are subtle. These findings emphasize the importance of considering tributary-specific attributes when assessing their role in sediment dynamics, as aggregated analyses can obscure localized effects. The study also identified methodological and contextual limitations, including the focus on surface sediment measurements and the unique geomorphic setting of the Dixence River. Future research should integrate subsurface sampling and temporal monitoring to capture the full complexity of sediment transport processes. Additionally, expanding the analysis to other dammed rivers with varying hydrological and geological conditions would enhance the generalizability of the findings. In conclusion, this work helps a bit in understanding of how dams and tributaries interact to shape sediment composition in Grade Dixence river and alpine rivers in general. The results show the potential of tributaries and so a more in depth and more detailed research is needed which will help in creating more efficient and sustainable sediment management strategies.

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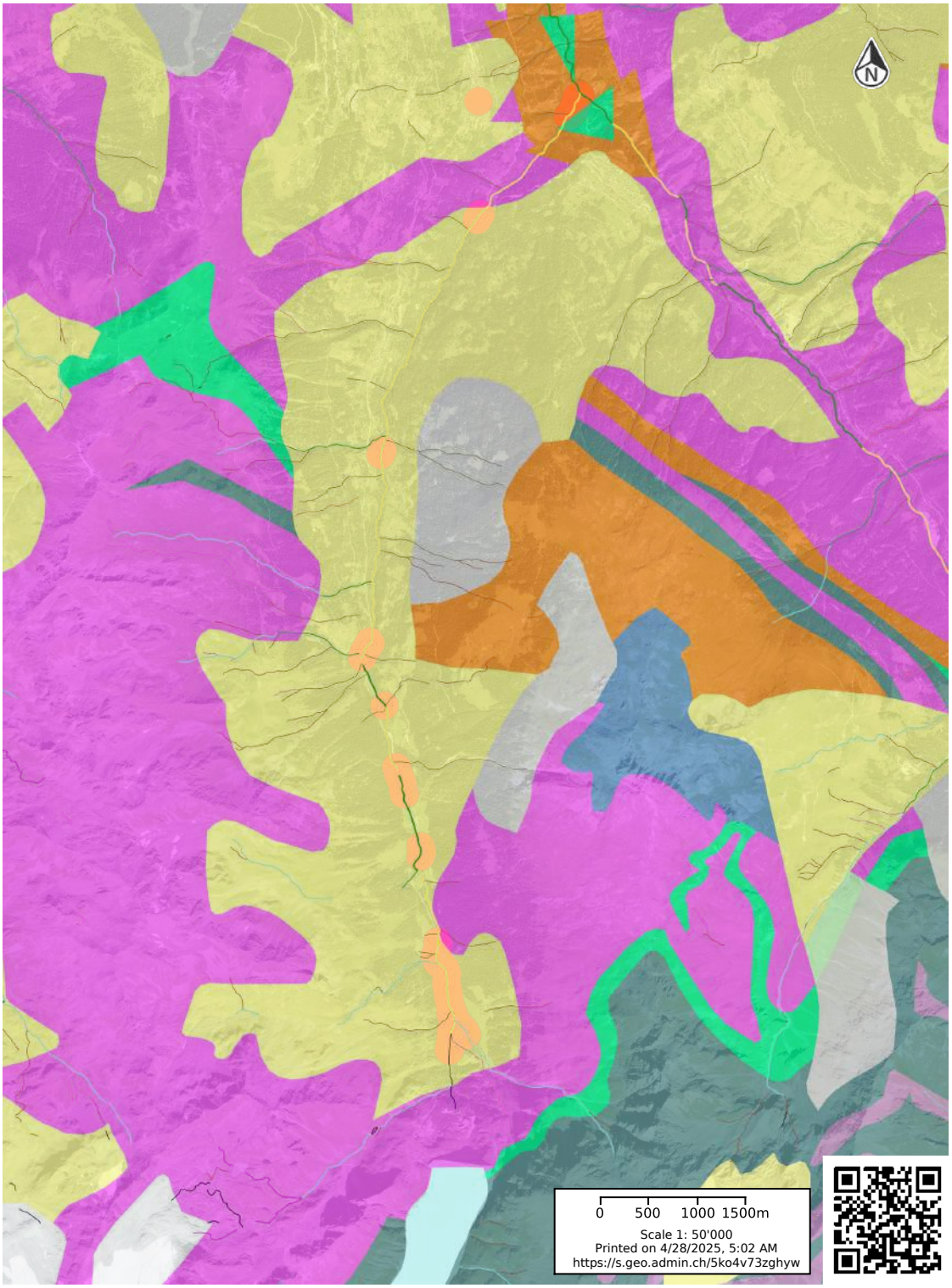
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Appendix A - Lithology Map



Legend
Lithology 500

- Clay, silts, sabbie
- Sands, gravels, cobbles
- Blocks, shingle
- Marls, partly sandstones
- Sandstones, partly marls
- Conglomerates, breccias
- Argillites, argillaceous shales
- Limestones, partly marls
- Dolomites
- Radiolarites, flint
- Argillites, evaporites, cornieules
- Granites, syenites
- Porphyrites, quartzitic porphyrites
- Volcanic rocks
- Marly shales, calcareous phyllites
- Calcareous and dolomitic marbles
- Quartzites
- Quartzitic phyllites
- Mica shists, gneiss
- Gneiss
- Amphibolites
- Basic rocks
- Ultrabasic rocks
- Water, lakes
- Glacier, firn

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